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Comparison of Captopril and Enalapril in the Treatment of Hypertension in Patients with Non-insulin Dependent Diabetes Mellitus and Nephropathy

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Eighteen patients with non-insulin dependent diabetes mellitus (NIDDM), hypertension and nephropathy were randomized to receive captopril or enalapril for 6 months. Two patients with serum creatinine of $> 400\,\mu$mol/l had to be excluded from the study because of rapidly deteriorating renal function after starting treatment. Of the remaining patients, 7 received captopril and 9 received enalapril. Blood pressure control was achieved in about 50% of patients with either drug alone. Serum creatinine and creatinine clearance were unchanged in both groups but there was a greater tendency for the former to increase in patients with higher pretreatment values. Proteinuria was reduced at 1 month only in the enalapril group which also showed a significant elevation of serum potassium after treatment. Captopril and enalapril have only a modest antihypertensive action in patients with NIDDM and nephropathy. Their use in patients with renal insufficiency must be balanced against the risk of further aggravating the deterioration of renal function.

Introduction

The control of systemic hypertension retards the progression of renal failure in patients with diabetic nephropathy [1]. Recent data obtained from animal experiments suggested that angiotensin converting enzyme inhibitors (ACEI) may be the drug of choice because they may selectively reduce intraglomerular hypertension which is thought to initiate and contribute to the relentless progression of diabetic nephropathy [2]. Several ACEI are currently available. Captopril is an ACEI with a short duration of action and has been shown to reduce proteinuria [3] and retard the progression of renal failure in patients with insulin dependent diabetes mellitus (IDDM) and nephropathy [4]. The use of the longer acting ACEI, e.g. enalapril or lisinopril, in the treatment of diabetic nephropathy in man has so far been only reported in a small number of patients [5]. Since the relative efficacy and safety of these two groups of drugs in the treatment of medical conditions known to benefit from the use of ACEI may be different [6], we conducted a randomized controlled clinical trial in which we directly compared the clinical and biochemical responses to captopril and enalapril in a group of non-insulin dependent diabetic (NIDDM) patients with co-existing hypertension and nephropathy.

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Patients and methods

All patients with NIDDM who had documented hypertension (resting sitting blood pressure >140/90) and positive urinary albustix were eligible for study. Two weeks before commencement of the study, all antihypertensive medications were ceased. All other medications including antidiabetic drugs were continued at the usual dosage. After randomization to receive either captopril or enalapril, the patient was admitted to the metabolic ward. Resting supine blood pressure was measured using a random zero sphygmomanometer with standard 14 cm size cuff after 30 min recumbency and the mean of three measurements was recorded. Mean arterial pressure was taken as the diastolic pressure plus one third of the difference between the systolic and diastolic blood pressure. The diastolic blood pressure was taken at the disappearance (phase V) of the Korotkoff sounds. Fasting blood was taken for the measurement of sodium, potassium, chloride, bicarbonate, urea, creatinine, urate, albumin, cholesterol, glucose, glycosylated haemoglobin, peripheral renin activity, aldosterone and osmolality. A spot urine sample was taken for the measurement of osmolality. The patient was given 300 ml of water to drink and urine was collected over 2 hours for the measurement of urinary N-acetyl glucosaminidase (NAG) and creatinine. Ammonium chloride at a dose of 0.1 g/kg was then given orally and the patient was instructed to drink 100 ml of water hourly for the next 6 hours. Urine was collected hourly for the measurement of pH, bicarbonate, titratable acid and ammonia. At the end of the sixth hour, arterial blood was taken for blood gas study. The patient was then instructed to collect a 12-hour urinary specimen at home for the measurement of sodium, potassium and creatinine. Twenty-four-hour urine was also collected on the day prior to admission for measurement of creatinine clearance and protein.

The starting dose of captopril was 50 mg daily in 2 or 3 divided doses and that of enalapril was 20 mg in 2 divided doses. The patient was then followed biweekly for 1 month and monthly for an additional 5 months. During the first 2 months of follow-up, the dose of ACEI was adjusted to achieve blood pressure control. If blood pressure control was not possible with the maximal dose of captopril (150 mg daily) or enalapril (40 mg daily), nifedipine and other antihypertensive drugs were added. During each follow-up visit, resting supine blood pressure, fasting blood for electrolytes, urea, creatinine, urate, albumin, cholesterol, glucose and glycosylated haemoglobin, and a 24-hour urine for creatinine and protein were measured. All the blood and urinary biochemical tests which had been done before treatment were repeated 2 months after the commencement of therapy.

Routine laboratory techniques were used for measuring serum and urinary electrolytes, creatinine, protein and osmolality and for measuring serum urea, urate, albumin, cholesterol, glucose and glycosylated haemoglobin. Plasma renin activity and plasma aldosterone were measured by radioimmunoassay using commercially available kits (Biodata, Milano, Italy). Urinary N-acetyl glucosaminidase (NAG) [7] and urinary pH, bicarbonate, titratable acid and ammonia [8]

were measured using previously described methods. All serum, plasma or urine samples not assayed immediately were stored at −20 °C until use.

For statistical analysis, Student's *t*-test was used, paired when appropriate. Linear regression was by the method of least squares. The values when skewed were log-transformed before comparison. A p-value < 0.05 was considered as significant.

Results

Patient characteristics

A total of 18 patients were recruited for the study. Eight were randomized to receive captopril and 10 received enalapril. Two patients, one from each treatment group, were withdrawn prematurely from the study because of rapidly deteriorating renal function after starting treatment. One was a 56-year-old male randomized to receive captopril and the other was a 60-year-old female who received enalapril. Both patients were on oral glibenclamide and had hypertension and advanced diabetic nephropathy at the time of entry into the study. Near normalization of blood pressure was achieved in both patients with either captopril or enalapril but a progressive rise in serum creatinine (from 405 to 501 μmol/l in the captopril-treated patient and from 412 to 834 μmol/l in the enalapril-treated patient) over a 2-month period resulted in both patients being withdrawn from the trial. Both patients had nephrotic range proteinuria but only in the enalapril-treated patient was proteinuria reduced (from 2.95 to 1.8 g/day). Furthermore, only the patient treated with enalapril showed a significant rise in her serum potassium level (from 4.1 to 5.7 mmol/l). Renal arteriogram was done in both patients and failed to reveal renal artery stenosis.

The clinical characteristics and baseline parameters of the remaining 16 patients are summarized in Table 1. All patients had serum creatinine less than 200 μmol/l (range 40–192). All were on oral hypoglycaemic drugs except one in the enalapril group who received insulin. There was no significant difference between the two groups with respect to any of the parameters measured but the urinary protein excretion in the enalapril group was higher than in the captopril group.

Blood pressure control

All except 1 of the 16 patients received ACEI as the only antihypertensive drug for the first 2 months of the study. The remaining patient had severe hypertension after his previous antihypertensive drugs were ceased during the run in period and required maximal dose of enalapril plus nifedipine and frusemide for adequate control. From the second month till the end of the study, 3 patients (43%) in the captopril group and 4 additional patients (55%) in the enalapril group required nifedipine for control of their blood pressure. The mean daily dose of

Table 1

Baseline clinical and biochemical characteristics of patients randomized to receive either captopril or enalapril

	Normal range	Captopril (n = 7)	Enalapril (n = 9)
Age (yrs)		51.9±7.0[a]	55.0±9.3
Sex ratio (M : F)		4 : 3	3 : 6
Duration of diabetes (yrs)		6.3±4.1	8.3±6.2
Mean arterial pressure (mm Hg)	70–107	127±11	125±11
Fasting blood glucose (mmol/l)	3.9–5.7	10.7±3.2	10.1±2.3
Glycosylated haemoglobin (%)	< 8.0	11.7±2.1	11.7±2.3
Serum sodium (mmol/l)	138–146	139±3	142±4
Serum potassium (mmol/l)	3.3–5.3	4.2±0.6	4.1±0.6
Serum albumin (g/l)	45–57	43±5	42±5
Serum urate (mmol/l)	0.16–0.47	0.31±0.13	0.35±0.08
Serum cholesterol (mmol/l)	2.9–5.6	5.4±0.8	5.7±0.9
Serum urea (mmol/l)	2.9–7.2	5.1±1.8	7.9±3.6
Serum creatinine (μmol/l)	60–120	82±27	98±44
Creatinine clearance (ml/min)	57–124	108±40	82±40
Urine/plasma osmolality ratio	1.3–3.6	2.0±0.5	1.8±0.7
Urine NAG/creatinine ratio	< 1.0	2.5±0.9	3.3±1.6
12-hour urinary sodium (mmol/12 h)	35–140	62±13	84±61
12-hour urinary potassium (mmol/12 h)	8–22	18±3	18±12
24-hour urine protein (g/day)	< 0.15	2.4±4.8[b]	4.0±3.6[b]
Plasma renin activity (ng/ml/h)	0.51–2.64	1.23±0.46	1.49±0.51
Plasma aldosterone (pg/ml)	12–150	100±65	60±30

[a] All values are expressed as mean ± standard deviation

[b] The 24-hour urinary protein values are skewed. There is no significant difference between the captopril and the enalapril group after log transformation

captopril at the second and sixth month was 71 ± 27 and 93 ± 45 mg and the corresponding mean daily dose of enalapril was 33 ± 20 mg and 38 ± 19 mg, respectively. The effect of captopril and enalapril on blood pressure control is shown in Fig. 1. Both captopril and enalapril significantly reduced the mean arterial pressure. Despite an increased dose of ACEI and the addition of nifedipine, the blood pressure in some patients escaped control from 3 months onwards. This trend was particularly obvious in the enalapril group. When the data of the two groups were pooled, the percentage change of blood pressure at 2 and

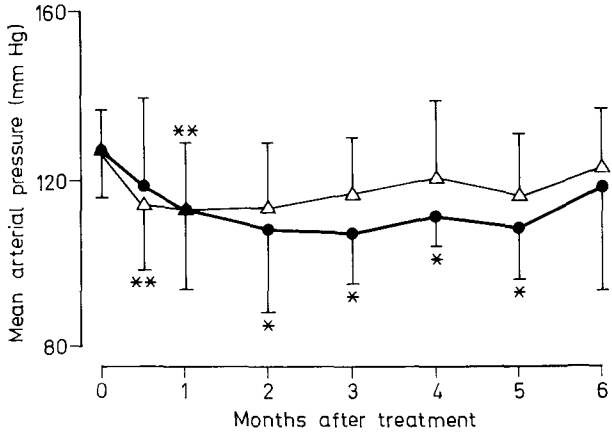

Fig. 1. Effect of capropril (●) and enalapril (△) on mean arterial blood pressure in non-insulin dependent diabetic patients with hypertension and nephropathy. * and ** indicate values significantly different from baseline ($p < 0.05$) in the captopril and the enalapril treated group

6 months after treatment directly correlated with the pretreatment proteinuria ($r = +0.51$, $p < 0.05$ and $r = +0.52$, $p < 0.05$ respectively; $n = 16$) but not with the pretreatment serum creatinine level.

Glycaemic control

During the 6 months of study, the dose of oral hypoglycaemic drugs was increased in 2 of 7 patients in the captopril group and in 2 of 9 patients in the enalapril group. The dose of insulin in the only patient receiving this treatment was not altered. Fasting blood sugar level progressively decreased from the baseline levels in both groups of patients (10.7 ± 3.2 to 8.2 ± 3.9 mmol/l and 10.1 ± 2.3 to 8.2 ± 3.8 mmol/l at 6 months, respectively, in the captopril and the enalapril group). The level of glycosylated haemoglobin was slightly reduced following treatment only in the enalapril group (11.8 ± 2.4 to 10.6 ± 2.4 at 6 months). A similar trend was also observed in patients on a constant dose of oral hypoglycaemic drugs.

Effect on renal function and proteinuria

There was no significant change in the serum creatinine and creatinine clearance at any point of time following treatment in either group. When the data of the two groups were pooled, the percentage change of serum creatinine at 6 months after treatment directly correlated with pretreatment serum creatinine level ($r = +0.62$, $p < 0.02$; $n = 16$), but not with the percentage change of

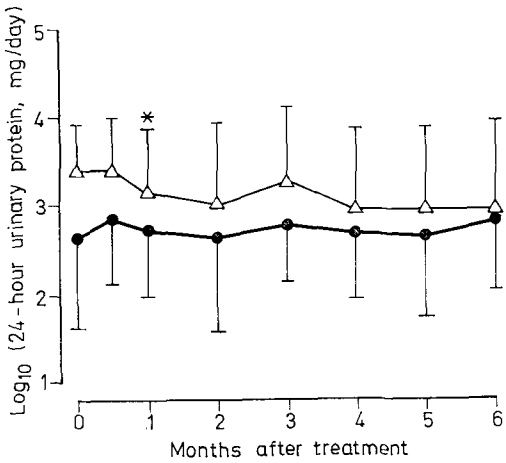

Fig. 2. Effect of captopril (●) and enalapril (△) on 24-hour urinary protein excretion in non-insulin dependent diabetic patients with hypertension and nephropathy. * indicates values significantly different from baseline ($p < 0.05$) in the enalapril treated group

blood pressure. The serum urea level was significantly higher than the pretreatment value only in the captopril group at 6 months ($p < 0.02$).

A significant reduction in the degree of proteinuria was demonstrated only in the enalapril group at 1 month (Fig. 2). The percentage change in proteinuria did not correlate with the change in blood pressure or creatinine clearance in either group.

Effect on other serum and urinary biochemical parameters

Serum sodium, chloride, bicarbonate, albumin, urate and cholesterol levels did not change significantly following treatment with either captopril or enalapril (Table 2). Although the serum potassium level progressively increased in both groups of patients, the difference from baseline was statistically significant only in the enalapril group ($p < 0.05$ at all points of time after treatment except for the value at 1 month) (Table 2). The percentage change in serum potassium correlated directly with the pretreatment serum creatinine level ($r = 0.65$ to 0.75, $p < 0.02$ to < 0.001 at 1–6 months) and inversely with the creatinine clearance ($r = -0.51$ to -0.63, $p < 0.05$ to < 0.01 at 1–6 months) but did not correlate with the change in fractional excretion of sodium or potassium. There was also no significant change in the latter two parameters after treatment (Table 2). Indexes of tubular function including spot urinary/serum osmolality ratio, urinary NAG/creatinine ratio and urinary acidification following ammonium chloride loading were similarly unaffected. Peripheral plasma renin activities were within

Table 2

Effect of captopril and enalapril on serum electrolytes, albumin, urate, cholesterol, plasma renin activity, aldosterone and urinary indexes of tubular function in NIDDM patients with hypertension and nephropathy

	Captopril (n = 7)		Enalapril (n = 9)	
	before	after*	before	after*
Serum sodium (mmol/l)	139±3	141±2	142±4	142±3
Serum potassium (mmol/l)	4.2±0.6	4.4±0.8	4.1±0.6	4.8±0.5 **
Serum chloride (mmol/l)	104±3	105±1	103±2	105±2
Serum bicarbonate (mmol/l)	28±2	27±3	26±3	27±3
Serum albumin (g/l)	43±5	42±2	42±5	41±4
Serum urate (mmol/l)	0.31±0.13	0.32±0.09	0.35±0.08	0.38±0.07
Serum cholesterol (mmol/l)	5.4±0.8	5.5±1.0	5.7±0.9	6.0±1.0
Plasma renin activity (ng/ml/h)	1.2±0.5	2.9±2.5	1.5±0.5	1.1±0.2
Plasma aldosterone (pg/ml)	100±65	92±51	61±30	68±31
Urinary 12-hour sodium (mmol)	62±13	78±24	84±61	80±78
Urinary 12-hour potassium (mmol)	18±3	22±6	18±12	17±17
Fractional excretion of sodium (%)	0.68±0.26	0.90±0.37	0.92±0.64	0.90±0.43
Fractional excretion of potassium (%)	6.0±0.9	6.8±4.3	9.7±5.6	8.7±6.3
Urinary serum osmolality ratio	2.0±0.5	2.4±0.4	1.8±0.7	1.9±0.6
Urinary NAG/creatinine ratio	2.5±0.9	2.1±1.1	3.3±1.6	2.7±1.5
6-hour urinary pH after acid loading	5.7±0.7	5.4±0.3	5.2±0.3	5.2±0.2

* Only 2-month posttreatment results are shown.
** $p < 0.05$ versus pretreatment values.

the normal range in the study population and failed to rise significantly after ACE inhibition. Plasma aldosterone levels also did not change significantly following treatment (Table 2).

Discussion

The present study shows that captopril or enalapril alone reduced the blood pressure in only half our patients. Despite the addition of nifedipine, the blood pressure in the other half remained difficult to control. The relative ease in the control of hypertension appears to be inversely related to the severity of the underlying nephropathy as the percentage change in blood pressure correlated directly with the pretreatment proteinuria. This would account for the greater

difficulty in controlling the blood pressure in the enalapril group as they had a higher pretreatment proteinuria when compared to the captopril group. This may also explain the discrepancy of the results of the present study compared to the previous ones which have reported satisfactory control of hypertension with ACEI monotherapy in patients with diabetes with early or no documented nephropathy [4, 9]. Recent studies have shown that diuretics are frequently required in addition to ACEI to achieve optimal blood pressure control in diabetic patients with more advanced nephropathy [5, 10], probably because in advanced diabetic nephropathy hypertension is more dependent on volume than on renin [11].

An earlier claim that the use of ACEI is safe in azotaemic diabetic patients with advanced nephropathy [3] is not confirmed by the present study. Two patients, one from each group, with a serum creatinine higher than 400 μmol/l had to be withdrawn from the study because of rapid deterioration in renal function following treatment. In both, renal artery stenosis was excluded by angiograms. Although the use of both captopril and enalapril in our patients with normal and mildly impaired renal function appeared to be safe, the greater rise of serum creatinine in patients with higher pretreatment serum creatinine level suggests that they should be used with caution even in the latter group of patients.

In laboratory animals with experimentally induced diabetes mellitus, there is a close association between intraglomerular hypertension, proteinuria and the development of glomerular sclerosis [2]. Therefore, it is generally felt that reduction in proteinuria is a desirable therapeutic objective. ACEI has been shown to reduce proteinuria in patients with IDDM with early or advanced nephropathy independent of its effect on blood pressure and glomerular filtration rate [3, 4], but recent studies have failed to confirm these results in patients with NIDDM [5, 10]. We have also found that captopril and enalapril have little effect in reducing proteinuria in patients with NIDDM. Although proteinuria was reduced in the enalapril group, the reduction was modest and not sustained and was only observed at 1 month after treatment. No change was noted in the captopril group.

Like other investigators [5], we have found that the serum potassium level increased significantly, especially in patients with impaired renal function, after treatment with enalapril. The change was less marked in the captopril group. The explanation for the rise in serum potassium remains obscure as fractional excretion of potassium, urinary acidification and serum bicarbonate did not change significantly after treatment.

When we planned our study, we were afraid that ACE inhibition would aggravate the high prevalence of hyporenin hypoaldosteronism in diabetic patients with hypertension and nephropathy [12]. However, in our study population, baseline plasma renin activity and aldosterone level were both within the normal range. Plasma aldosterone level was not significantly altered after ACE inhibition, but this may be attributed to the rise in serum potassium level in our patients. In contrast to previous reports [4, 9, 10], plasma renin activity did not change significantly following ACE inhibition. As some of the posttreatment plasma

samples had been stored for longer than 6 months before assay, the lack of rise of plasma renin after treatment may be due to a loss of renin activity after prolonged cold storage. As far as we could determine, this was not due to the lack of drug compliance in our patients.

To the best of our knowledge, this study is the first to compare the relative efficacy of captopril and enalapril in the treatment of hypertension in patients with NIDDM and nephropathy. We conclude that both drugs have similar effects on blood pressure, renal function and proteinuria but enalapril induces a more marked elevation of serum potassium especially in patients with renal insufficiency. In the latter group of patients, ACEI should be used with caution as they may further accelerate the deterioration of renal function.

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